

## SOLA STAVANGER, Norway

WMO: 014150

Lat: 58.884N

Lon: 5.637E

Elev: 29

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 16.2       | 20.3 | 6.6                           | 7.7 | 22.4 | 10.7 | 9.5 | 24.2 | 32.7                  | 38.3 | 30.0 | 38.6 | 3.2                   | 140  | 0.675 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 10.4                   | 75.7            | 63.2 | 71.9 | 61.6 | 68.7 | 60.7 | 65.4                | 72.5 | 63.5 | 69.6 | 61.9 | 67.2 | 11.4                 | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 62.7                            | 85.3 | 68.2 | 60.9 | 80.1 | 66.3 | 59.3 | 75.5 | 64.8 | 30.3          | 72.5 | 28.9 | 69.5 | 27.7 | 66.9 | 74.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 29.1          | 25.5 | 22.7 | 12.7                       | 81.7 | 4.7                | 3.7 | 9.3                                                | 84.4 | 6.6        | 86.6 | 3.9        | 88.7 | 0.5        | 91.4 |    |    |
| (5)               |      |      | 12.0                       | 67.8 | 4.9                | 2.8 | 8.5                                                | 69.8 | 5.6        | 71.4 | 2.9        | 73.0 | -0.6       | 75.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 47.0   | 35.8  | 35.5  | 38.3  | 44.4  | 50.1  | 55.1  | 59.8  | 60.3  | 55.8  | 48.6  | 41.8  | 37.4 |
|                                                                               | DBStd       | 10.51  | 6.63  | 6.18  | 5.14  | 5.24  | 5.52  | 4.50  | 4.78  | 4.59  | 4.42  | 5.08  | 6.20  | 7.44 |
|                                                                               | HDD50       | 2189   | 440   | 406   | 364   | 183   | 63    | 4     | 0     | 0     | 5     | 85    | 248   | 391  |
|                                                                               | HDD65       | 6618   | 905   | 826   | 828   | 619   | 464   | 301   | 176   | 160   | 278   | 510   | 695   | 856  |
|                                                                               | CDD50       | 1086   | 0     | 0     | 0     | 14    | 66    | 157   | 305   | 320   | 180   | 40    | 4     | 0    |
|                                                                               | CDD65       | 37     | 0     | 0     | 0     | 0     | 1     | 3     | 15    | 16    | 2     | 0     | 0     | 0    |
|                                                                               | CDH74       | 185    | 0     | 0     | 0     | 0     | 9     | 20    | 87    | 63    | 6     | 0     | 0     | 0    |
|                                                                               | CDH80       | 20     | 0     | 0     | 0     | 0     | 0     | 1     | 13    | 6     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 10.2   | 11.6  | 10.6  | 10.4  | 9.9   | 10.0  | 9.9   | 8.9   | 9.2   | 9.7   | 10.7  | 10.3  | 10.6 |
| Precipitation                                                                 | PrecAvg     | 47.2   | 4.1   | 3.0   | 2.8   | 2.2   | 2.6   | 2.7   | 3.5   | 4.8   | 5.6   | 5.7   | 5.5   | 4.7  |
|                                                                               | PrecMax     | 68.2   | 9.3   | 7.0   | 7.9   | 5.0   | 6.6   | 6.3   | 9.6   | 9.5   | 12.9  | 11.8  | 9.6   | 9.8  |
|                                                                               | PrecMin     | 33.1   | 0.2   | 0.0   | 0.1   | 0.1   | 0.1   | 0.6   | 0.8   | 0.2   | 1.1   | 1.5   | 1.0   | 0.3  |
|                                                                               | PrecStd     | 7.6    | 2.4   | 1.7   | 1.8   | 1.0   | 1.4   | 1.3   | 2.0   | 2.0   | 2.2   | 2.4   | 2.2   | 2.4  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.6  | 48.0  | 54.5  | 66.8  | 74.2  | 77.0  | 81.8  | 80.1  | 73.7  | 63.2  | 56.0  | 51.1 |
|                                                                               |             | MCWB   | 46.7  | 45.7  | 46.2  | 52.5  | 59.1  | 59.9  | 65.1  | 65.2  | 62.1  | 56.5  | 51.7  | 48.7 |
|                                                                               | 2%          | DB     | 46.7  | 46.3  | 48.9  | 60.4  | 68.3  | 70.6  | 75.9  | 75.1  | 68.2  | 59.3  | 53.5  | 49.5 |
|                                                                               |             | MCWB   | 44.9  | 44.8  | 44.1  | 49.3  | 56.1  | 59.2  | 64.1  | 64.0  | 59.3  | 54.6  | 50.2  | 47.5 |
|                                                                               | 5%          | DB     | 45.5  | 44.9  | 46.5  | 56.3  | 63.4  | 66.0  | 71.5  | 71.3  | 64.8  | 57.4  | 51.7  | 48.0 |
|                                                                               |             | MCWB   | 43.7  | 43.5  | 43.1  | 47.9  | 54.1  | 57.7  | 62.4  | 62.7  | 58.1  | 53.4  | 49.1  | 46.0 |
|                                                                               | 10%         | DB     | 44.4  | 43.9  | 44.9  | 52.9  | 59.2  | 62.5  | 67.9  | 68.1  | 62.7  | 55.6  | 50.2  | 46.7 |
|                                                                               |             | MCWB   | 42.6  | 42.4  | 42.2  | 46.4  | 51.9  | 56.4  | 61.5  | 61.6  | 57.2  | 52.0  | 47.7  | 44.7 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 47.2  | 46.4  | 47.9  | 54.8  | 61.7  | 63.3  | 67.8  | 68.2  | 63.5  | 58.0  | 53.2  | 49.5 |
|                                                                               |             | MCDB   | 47.9  | 46.9  | 50.9  | 62.9  | 71.1  | 69.7  | 76.6  | 73.6  | 69.3  | 60.8  | 54.8  | 50.2 |
|                                                                               | 2%          | WB     | 45.4  | 45.1  | 45.4  | 51.3  | 57.5  | 60.8  | 65.6  | 65.6  | 61.2  | 55.8  | 51.1  | 47.9 |
|                                                                               |             | MCDB   | 46.2  | 45.8  | 47.7  | 57.8  | 65.5  | 67.6  | 73.2  | 71.9  | 65.9  | 58.4  | 52.9  | 48.9 |
|                                                                               | 5%          | WB     | 44.2  | 44.0  | 44.0  | 49.2  | 54.9  | 58.8  | 63.7  | 63.8  | 59.5  | 54.5  | 49.6  | 46.5 |
|                                                                               |             | MCDB   | 45.1  | 44.8  | 45.7  | 54.6  | 61.7  | 64.9  | 69.9  | 69.8  | 63.8  | 56.8  | 51.1  | 47.6 |
|                                                                               | 10%         | WB     | 42.8  | 42.6  | 42.8  | 47.4  | 52.6  | 57.0  | 61.9  | 62.1  | 58.0  | 52.9  | 48.1  | 45.0 |
|                                                                               |             | MCDB   | 44.0  | 43.7  | 44.5  | 51.8  | 58.3  | 61.8  | 67.2  | 67.1  | 61.8  | 55.3  | 49.8  | 46.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 7.9   | 8.5   | 9.5   | 11.2  | 11.4  | 10.3  | 10.2  | 10.4  | 9.9   | 9.3   | 8.5   | 8.2  |
|                                                                               |             | MCDBR  | 7.3   | 6.9   | 10.8  | 17.2  | 18.9  | 17.0  | 17.5  | 15.5  | 13.2  | 10.4  | 8.2   | 6.9  |
|                                                                               |             | MCWBR  | 7.2   | 6.5   | 7.8   | 9.7   | 10.0  | 9.2   | 8.7   | 8.1   | 7.7   | 7.6   | 7.6   | 7.2  |
|                                                                               | 5% WB       | MCDBR  | 7.2   | 6.4   | 9.2   | 14.3  | 17.0  | 14.8  | 15.1  | 13.5  | 11.4  | 8.5   | 7.8   | 6.9  |
|                                                                               |             | MCWBR  | 7.5   | 6.4   | 7.5   | 8.9   | 10.0  | 8.8   | 8.4   | 8.0   | 8.0   | 7.3   | 8.0   | 7.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.309  | 0.332 | 0.352 | 0.383 | 0.372 | 0.365 | 0.379 | 0.380 | 0.361 | 0.341 | 0.313 | 0.287 |      |
|                                                                               | taud        | 2.177  | 2.278 | 2.320 | 2.291 | 2.405 | 2.454 | 2.434 | 2.430 | 2.468 | 2.410 | 2.256 | 2.126 |      |
|                                                                               | Ebn at Noon | 154    | 215   | 248   | 260   | 273   | 276   | 269   | 259   | 246   | 212   | 153   | 118   |      |
|                                                                               | Edh at Noon | 15     | 23    | 29    | 35    | 33    | 32    | 32    | 30    | 25    | 20    | 14    | 11    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 94     | 285   | 673   | 1213  | 1640  | 1764  | 1567  | 1210  | 777   | 372   | 131   | 61    |      |
|                                                                               | RadStd      | 15     | 48    | 78    | 119   | 180   | 196   | 155   | 102   | 68    | 44    | 19    | 7     |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -247  | N/A   | N/A   |
| (47) Regional (30 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -209  | N/A   | N/A   |

Nomenclature: See separate page